

MULTI-AXIS ENDPOINTING, FEATURE DISCRIMINATION, COMBINED ULTRASOUND + CONTACT CONFIRMATION, AND ROBUSTNESS ACROSS USERS (NON-LIMITING)

A. Overview (Non-Limiting)

In various embodiments, Node 4 uses ultrasound not only to estimate a single endpoint distance, but to support robust endpoint discrimination and improved repeatability across variable internal conditions by leveraging multi-axis interrogation, redundant confirmation sensors, and consistency constraints.

B. Multi-Axis / Multi-Angle Interrogation (Non-Limiting)

In various embodiments, Node 4 includes one or more ultrasound elements oriented to interrogate tissue along different angles or axes. Non-limiting configurations include: 1. a primary longitudinal transducer oriented toward an endpoint direction; 2. one or more offset transducers oriented at a slight angle relative to the longitudinal axis; 3. a small array enabling beam steering or directional sensitivity; and 4. an acoustic lens or shaping element that biases sensitivity toward a target region. In certain embodiments, multi-angle interrogation reduces specular dependence and improves endpoint stability by selecting the most consistent reflector across angles.

C. Endpoint Feature Discrimination (Non-Limiting)

In various embodiments, ultrasound echoes may be used to discriminate among multiple potential internal reflectors. Non-limiting discrimination methods include: 1. selecting reflectors based on persistence across pulses and angles; 2. using echo envelope shape and peak sharpness as a reflector signature; 3. tracking a reflector using continuity constraints (temporal tracking) to prevent jumping between surfaces; and 4. optionally applying a classifier that maps echo features to an endpoint-likelihood score. In certain embodiments, the system outputs an endpoint distance only when an endpoint-likelihood score exceeds a threshold.

D. Combined Ultrasound + Contact Confirmation (Non-Limiting)

In various embodiments, ultrasound endpointing is paired with one or more contact confirmation sensors to increase reliability and reduce spoofing/false readings. Non-limiting combinations include: 1. capacitive proximity/contact confirmation electrodes indicating tissue contact; 2. circumferential pressure sensing (FSR/capacitive ring or film band) confirming radial contact and providing contact uniformity indices; 3. temperature context confirming plausible coupling and reducing drift; and 4. IMU gating to suppress measurements under motion. In certain embodiments, the system requires a contact confirmation condition before reporting ultrasound distance.

E. Robustness Across Moisture, Tissue Variability, and Orientation (Non-Limiting)

In various embodiments, Node 4 is configured to operate across variability in moisture, tissue compliance, and orientation by: 1. adjusting transmit amplitude/pulse repetition based on coupling quality indices; 2. using multi-pulse averaging and stability thresholds; 3. using orientation-aware gating (IMU) and position stability constraints; and 4. using repeated measurements to compute a repeatability score. In certain embodiments, the system provides a guided positioning mode and displays a stable coupling indicator prior to recording a measurement.

F. Repeatability Scoring and Session Outputs (Non-Limiting)

In various embodiments, Node 4 computes repeatability and robustness metrics from ultrasound endpoint outputs, including: 1. intra-session stability (standard deviation or spread of endpoint estimates); 2. inter-session repeatability (agreement across sessions under matched context); 3. coupling stability indices; and 4. confidence indicators derived from the above. In certain embodiments, reported outputs include a distance index plus a repeatability/confidence label.

G. Optional Use with Controlled Radial Stimulus (Non-Limiting)

In various embodiments, ultrasound endpointing is performed during controlled radial stimulus cycles (inflation/expansion). Non-limiting derived metrics include: 1. endpoint distance change vs stimulus level; 2. compliance indices from pressure vs displacement relationships; and 3. contact uniformity indices that validate stimulus symmetry. In certain embodiments, the system detects constraint-limited responses where pressure increases without corresponding displacement and labels those windows accordingly.

H. Variations

Any embodiment described herein may vary transducer count, orientation, algorithms, thresholds, and redundant confirmation sensors while maintaining the functional intent of robust ultrasound-based endpoint distance estimation with repeatability scoring and contact-confirmed validity.